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For the final project, you will design and conduct a multi-week project on an NLP topic of your choice.

You are required to typeset your midway report and final report in the Association for Computational Linguistics (ACL) paper format. You are highly encouraged to use LaTeX to do this, as essentially every NLP project published in the field is typeset with LaTeX. You are not required to use it, but it generally makes typing equations and formatting figures and tables much easier.

You can use this linked template to typeset your materials.<sup>1</sup> If you strongly prefer to use Word, follow this link to the official ACL Word template.<sup>2</sup>

This will be an empirical project where you will perform some original experiment(s) to get novel results. (Or, if you choose, you may aim to do a reproduction of prior work, although this should ideally introduce at least one new element.) As this a course project, this would be a good opportunity to potentially try out a risky idea, and/or to get feedback on your ideas!

## 1 Final Project Proposal (5 points; due Oct. 20)

First, you will come up with a project idea. The project proposal should be a 1–2-page paper (11-12 point text, but otherwise in any format) containing each of the following:

- State the specific NLP problem or task you will be working on. Why is it interesting or important? Formally define the task; you may find math or figures helpful for describing it.
- Perform a literature search and briefly describe relevant prior work. You don't need a full related work section, but you should mention the most relevant prior work you've found and state how your project idea relates to it.
- Provide an outline of your project. What **datasets**, pre-existing **models** (if applicable), and **software** you will use? What methods will you implement?

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<sup>1</sup><https://www.overleaf.com/latex/templates/association-for-computational-linguistics-acl-conference/jvxskxpznfj>

<sup>2</sup>[https://2023.aclweb.org/calls/style\\_and\\_formatting/](https://2023.aclweb.org/calls/style_and_formatting/)

- An experimental design. How will you know if you have effectively solved your problem? You should clearly identify your **evaluation** metrics, or propose a novel one if this is the focus of your project. Also describe at least one simple and easy-to-implement baseline you could compare against, like a pre-trained LM with no fine-tuning, an existing statistical method, or a very simple classifier.

I will provide feedback and guidance on directions that will help maximize the project's chances of succeeding. You will *not* be evaluated on how good the idea is; this is a stage for getting feedback and refining the idea.

The project proposal is a contract. If I give you full credit, I expect you to implement it and analyze the results, and will give you full credit for the final report if you do so (assuming you follow all formatting instructions and meet the clarity expectations). If you turn in a proposal that does not receive full credit, you may revise and resubmit it *within 1 week of receiving your project proposal grade* before moving forward. But the longer you take to define your project, the less time you will have to do it, so do the best you can before the proposal deadline.

One member of your team will upload your proposal as a PDF to Gradescope. Be sure to include a title and the names of all of your team members in the PDF, and to add your team members to your Gradescope submission.

## 2 Midway Report (15 points; due Nov. 12)

The midway report is a progress report of 2–4 pages. It must be in Association for Computational Linguistics format. It should be written in the style of a short research paper. Doing well on this part of the project will benefit you greatly, as it will mean you have less work to do to get a good grade on the final report. I will also be able to provide better feedback if your report is well-written.

The report should be written in such a way that another student in the class could read it using the content we've covered in class so far (this will become relevant in a moment). This means that if you're using methods or terminology outside of what we've covered in class, you will need to describe and/or define it. That said, you don't need to explain concepts we've covered in class, like what perplexity is. Because the page limit is relatively short, you will need to be concise and write clearly!

Be sure to include the following in your report:

1. **Abstract:** A concise summary of your report. This should be less than 250 words (but ideally even shorter). It should summarize the problem and highlight your main findings so far. I often find that it's easiest to write abstracts *last*, once I've written and thought through the content of the entire paper.
2. **Introduction:** This section introduces the reader to the problem area you're investigating. Give a specific example and a technical definition of the problem. Elaborate on the following:
  - Why is your problem interesting or important? If you could solve this problem, would you be able to answer a scientific question or solve a real-world problem? Would it improve the efficiency, performance, or usability of existing systems?
  - Why is the problem hard, and how do obvious solutions fail? For empirical projects, you could answer this question by implementing and/or running a simple and obvious baseline, and (if you have time) analyzing failure cases.
3. Description of progress. For empirical projects, include the following sections:
  - **Methods:** Describe the method you have implemented. It should be clear and give enough technical detail that someone who wasn't on your team could reimplement your approach and

get similar results. This could include the names of the LMs you based your method on (if applicable), the algorithm(s) you used, and possibly some formal equations or illustrations if helpful.

- **Experiments:** Describe the dataset(s) you're using and give citations for each. Also describe the evaluation metrics or methods you're using to measure how well your methods and baselines perform. Be sure to include any other important details like key hyperparameters, training time, how long your method took to run, etc.
  - **Results:** This is where you describe any quantitative or qualitative results that you've obtained so far. Use tables and plots; do not put numbers in the main text. Do your findings look like what you expected to see so far? Why or why not? If you've done some error analysis or qualitative analyses, what do they tell you about how your method could be improved?
4. **Project plan:** This is where you describe your plan for the rest of your project. Include a list of experiments or analyses. This should be clear, concrete, and reasonable given the time left before the final report deadline. It's totally fine if this deviates from your project proposal! Just note how your plans have changed and why.
5. **References:** These should ideally be compiled with BibTeX. Note that references do not count toward the page limit; you can include as many as you like. At the very least, be sure to include the papers you cited in your proposal (unless you have switched directions), as well as any model(s) dataset(s) you have used. Also cite the most related systems/metrics/ideas to those you are developing.

Grading will be based on the following criteria:

- **Level of progress:** Has there been reasonable progress made so far? Ideally, you should be at least halfway done with running experiments and/or reading papers by this point, but this isn't a hard rule. Good progress will depend on the kind of project you're pursuing, and I will take this into account when grading.
- **Quality of writing:** Are important technical details clearly presented? Is the motivation clear? Are the results or taxonomy presented in a way that makes them easy to find and easy to understand, even before the reader looks at the text? Is the requested information provided to an appropriate amount of detail (given the page limit)? Is everything in ACL format, and does it adhere to the page limit? Can other students in the course understand what you've written given what we've covered in class? You may use AI to help your writing, but beware: many papers I have reviewed at conferences have recently been rejected because they either (i) contained made-up references (typically hallucinated by LLMs), or (ii) were written in a verbose and overly dense/hard-to-read style characteristic of models like Claude.
- **Clear trajectory:** Do you have a **concrete** plan for the rest of the project? Is the plan reasonable given the timeframe?

We will be using OpenReview for the midway reports. You will upload an *anonymized* PDF to OpenReview at this portal. Be sure to use the "review" argument on Overleaf so that the author names *do not* show up in your submission PDF. Only one member of your team should upload the PDF to OpenReview; be sure to add your team members to your submission.

Your midway report will receive feedback from 2 anonymous peer reviewers as well as me (the "meta-reviewer", but also technically a reviewer). Note that your grade will not be directly dependent on your review scores, as these may vary significantly regardless of your work's quality or adherence to the criteria. I will assign a grade that reflects alignment with the midway report criteria.

### 3 Peer Review (5 points; due Nov. 20)

You will be assigned to anonymously review 2 midway reports in the class. You will submit your reviews on OpenReview. There are three main goals behind this exercise: (i) to show you how the peer review system works, (ii) to get you a bit of extra constructive feedback from perspectives that differ from mine, and (iii) to give you practice writing reviews for research papers(-in-progress).

Both of your reviews should do the following:

1. Start with a **summary** of the paper. What question are they asking, or what problem are they trying to solve? How are they going about investigating/addressing it? What experiments have they conducted? What do the results so far look like, and how do they relate to the authors' research questions? Note that this is not the place to give opinions; the authors of the report should mostly agree with your summary.
2. Provide a few **strengths** of the paper. What did you find interesting or impressive about the main question or the methods, and/or what did it do well in general? Are the experiments well-designed or particularly thorough? Is the paper clearly written? Is the plan for the remainder of the project concrete, sensible, and/or feasible in the time left in the course?
3. List any **weaknesses** you noticed. Are there certain conclusions that are not (yet) well-supported by the experimental evidence? Is the related work not sufficiently well-surveyed (e.g., is there a paper doing something very similar that wasn't cited)? Is the writing hard to understand?
4. List any specific **comments or suggestions** (e.g., suggest some way of reformatting a figure to make it clearer, or suggest a new experiment that would help the authors make their main point). Also feel free to ask the authors **questions**. They will not be able to directly respond, but the question may help the authors think through some key aspect of their project or presentation.

Above all else, let's be *constructive* and *respectful*! Your feedback should be aimed at helping the authors improve their paper in the time they have left, not tearing them down nor fully accepting everything in the paper.

Your review will be graded on the following criteria:

- Is the summary accurate?
- Do the strengths and weaknesses give a fair and reasonable picture of what was done well or what could be improved, given the time the authors had/will have to do the project? You are allowed to take a positive or negative (or ambivalent) overall stance if you choose, but your opinion should be backed up with clear arguments and evidence.
- Are there any concrete suggestions or questions that would help the authors?
- Is the tone of the review neutral and respectful?
- Is the review sufficiently but not overly detailed? As a heuristic, aim for about one paragraph of summary, about 3 strengths and 3 weaknesses, and at least 1 or 2 suggestions, comments, or questions. A longer review is not necessarily better, nor is an overly short one.

I will provide links to examples of high-quality reviews on the course website before you submit your midway reports.

## 4 Poster Presentation (PDF due Dec. 4; poster session on Dec. 10)

You will design a poster to present during our class poster session on the last day of class. Note that this grade is attached to the final report grade; if you submit a poster on time and are able to describe your work clearly, you will receive full points. Otherwise, penalties will be applied to the final report grade (up to a maximum of 10 points). It takes students (and professors!) many years to learn how to present well, so don't worry, these won't be picked apart in fine detail. Feel free to have some fun with the poster and practice your marketing/graphic design skills.

I'll need the PDF  $\approx$  1 week in advance so that I can get them printed in time for the poster session. Your poster should be in **portrait** orientation, and be no larger than 36" (H) x 24" (W). Here are some examples of good posters:

- You et al. (2025). "Probabilistic Soundness Guarantees in LLM Reasoning Chains." Actionable Interpretability Workshop @ ICML.
- Yao et al. (2025). "Both Direct and Indirect Evidence Contribute to Dative Alternation Preferences in Language Models." COLM. (Not the right size for our purposes, but note the light use of text and the focus on *visual* presentation of information.)
- Castro et al. (2019). "Towards Multimodal Sarcasm Detection (An *Obviously* Perfect Paper)." ACL.

You may use whatever software you like to make the poster. I personally use Keynote or Powerpoint and set the size of the slide to match the poster size.

## 5 Final Report (25 points; due Dec. 17)

This should extend your midway report, filling in new results and missing details. This should be 4–6 pages in ACL format. I will make expectations clear in comments on your midway report. In general, empirical projects should aim for at least a couple additional experiments or analyses over the midway report.

Grading will be based on similar criteria to the midway report. In addition to those criteria, I will also consider the following:

- The thoroughness of your experimentation
- The originality of your ideas and analysis
- The amount of thought, effort, and creativity put into your project
- The performance of your methods (if you are doing an empirical project)
- The clarity of your writing and analysis, and the presentation of your results
- Whether you incorporated the feedback you received on your proposal and midway report

After your conclusion, I ask that you include a brief summary of each team member's individual contributions to the project in a section called **Author Contributions**. This should be only 1–2 sentences per person, and will not count toward the page limit. I will generally give each team member the same grade, but if there is a significantly unequal distribution of effort, I may investigate further and potentially give different grades to different team members.

One member of your team should upload your PDF to Gradescope. This PDF should be like a "camera-ready", in that it should contain the names of all authors in the PDF.

## 6 General Tips and Project Ideas

To get a feel for the style of a good paper, read some of the research papers linked in the syllabus. None of them are perfect, so read attentively: what do you find helpful or not when reading these papers? Copy the style of papers you find readable or enjoyable, and avoid copying the style of papers that are unclear or difficult to read.

**How to write well.** Here are some great tips on writing good technical papers:

- Write the Paper First, by Jason Eisner
- Tips for Writing Technical Papers, by Jennifer Widom
- Ten Simple Rules for Mathematical Writing, by Dimitri Bertsekas

**Project ideas.** Here are some ideas for empirical projects:

- A modified GRPO that normalizes with a moving reward average instead of a group-wise average
- A manually curated preference dataset to reduce sycophancy in LLMs
- Steering neurons or directions in language model representations that encode Big Five personality features
- Training or developing a harness for a compute-efficient local agent
- Creating and implementing a new evaluation metric for some task/system
- Collecting human annotations to meta-evaluate an existing metric or interpretability method
- Extending one of the homework assignments. For example, you could propose an architectural modification to your LM from HW1, or some extension of your GRPO implementation from HW2
- Training a probe to detect some abstract concept in an LLM's activations

You have broad freedom to choose what you would like to work on, but here are some general characteristics of good projects:

1. Only requires the training of small LMs or building on existing larger-scale LLMs. If you want to train models, you should be able to use small models. If you want to design reasoning harnesses or prompt models, you should use the strongest models you can get access to (either frontier models if you're prompting, or the best open-source models you can use, like Qwen or Gemma 4).
2. Multiple success modes. You can define some initial goal which is likely achievable, as well as some stretch goals if you achieve it.
3. Enough analysis/experiments or easy to pivot: the project is not just an all-or-nothing success or failure. You could, in theory, provide an interesting analysis of why the results turn out the way they do either way.
4. Avoids high compute requirements: projects involving GRPO on large models, large datasets, etc. may not be feasible with the SCC unless you have access to your own compute resources

## 7 Late Policy

If you have any free late days remaining, you may use them for the project proposal or midway report. If you have none left, you will lose 10% of that milestone's grade for each late day. As long as you submit the proposal on time, proposal revisions do not count toward your free late day limit. Note that there will not be extensions on the midway report deadline even if your proposal takes a while to get right, so do aim to get this right the first time around!

Late submissions will not be allowed for the final report.

## 8 AI Policy

See the course website for more details. Long story short, you are allowed to use AI tools for your implementation and experiments, as well as for writing help, but you are fully responsible for what you submit. If your paper is found to contain false information, hallucinated references, is generally difficult for me or students to read, or if the provided code is found not to have the described functionality, this may result in significant point deductions.